

# Regression Utility of a 10,240-Byte Tabular Generator

The DOPE Project

**Abstract**—The displayed DOPE generator is a compact residual: at most 12 screened inputs, a width-16 tanh hidden layer, and a refit linear readout. AdamW trains the hidden basis for 2,048 steps at learning rate 0.002, with no embedding. Each method keeps its own selection objective. After selection, artifacts are scored by retention, the share of real-versus-null validation improvement recovered by an auditor trained on synthetic rows, and by charged bytes. On 97 paired PMLB regression lineages the CatBoost median is 0.940 (95% lineage-bootstrap interval 0.914 to 0.976), against 0.656 for a native-selected Gaussian copula, 0.586 for a study Chow–Liu tree, and  $-0.012$  for study independent marginals. Holm-adjusted Wilcoxon tests separate these three CatBoost comparisons. The linear auditor does not separate DOPE from the Gaussian copula. An unpooled 21-lineage block compares CTGAN and TVAE. An unpooled 6-lineage Forest-Flow block reaches similar CatBoost retention at 115,202,253 to 1,085,155,062 bytes. 98/100 displayed DOPE artifacts sit inside the 10,240-byte cap. Official tests were not opened. Master Fitness v2 is null. TabSyn, a 100-lineage TabDDPM panel, privacy attacks, fit-seed variance, and the full ablation grid are not measured.

**Index Terms**—Synthetic tabular data, retention, artifact bytes, paired tests.

## I. INTRODUCTION

A generator paper often reports the metric that also selected the configuration. Those native scores live on different scales: predictive efficacy for CTGAN and TVAE [1], held-out log density for a Gaussian copula [2], a forest density for ARF [3], and a diffusion or CART objective for other families [4], [5]. This paper separates selection from the common measurement. After each method has been selected on its own objective, the shared columns are retention and the bytes charged to the artifact.

The decision those columns can support is narrow. It is whether a network small enough to ship under a 10,240-byte cap still preserves downstream regression utility on real validation rows. It is not a declaration that one generator is best, and it is not a certification score. Master Fitness v2 (MFS-v2) is defined in Section III and is null on every method here, because the six components and the hard-gate map are not all present.

The contributions are:

- 1) The displayed residual and the sealed-test training pipeline (Section III, Fig. 1, (1)).
- 2) Retention and charged bytes with lineage-bootstrap intervals (Fig. 2, Tables IV and V).
- 3) Paired Wilcoxon tests with Holm correction, rank-biserial correlations, and a Friedman–Nemenyi diagram on the common lineage set (Fig. 3, Table IV).
- 4) A separate six-lineage Forest-Flow block (Table VII).

- 5) An explicit account of what this panel does not measure, including MFS-v2 (Table VIII and Section VIII).

An unexecuted method is not a win for DOPE. Density, neural, Forest-Flow, and BeyondArena blocks are not pooled.

## II. RELATED WORK

Tabular synthesis has moved from parametric copulas [2] and Chow–Liu trees [6] through conditional GANs and variational autoencoders [1] to diffusion and score-based models [4], [7]–[12]. Autoregressive and language-model generators [13], [14], prior-data fitted generators [15], [16], CTAB-GAN and CTAB-GAN+ [17], [18], and graphical neural generators [19] target the same mixed-type tables. ARF estimates density with a forest [3]. CART synthesis is the synthpop line [5]. A recent survey collects these design choices [20]. This paper runs the Gaussian copula, study Chow–Liu, study independent marginals, CTGAN, TVAE, and a 6-lineage Forest-Flow confirmation. TabSyn, TabDDPM on the 100-lineage panel, STaSy, CoDi, TabDiff, CTAB-GAN+, GReaT, REaLTabFormer, and TabPFGen are cited and not measured. SMOTE [21] was not run. This panel scores regression retention, not classification oversampling.

Differentially private Bayesian networks and marginal synthesizers [22]–[24] optimize a privacy budget. Membership inference and shadow-model attacks are a separate detection literature [25]–[27]. Formal differential privacy is a different claim from an empirical resemblance score [28], [29]. Utility reviews separate marginal fidelity from downstream prediction [30], [31]. Alpha-precision, beta-recall, SDMetrics, and synthcity are the usual fidelity reports [32]–[34]. The density cells already store a classifier two-sample statistic, a mean marginal Kolmogorov–Smirnov distance, and a pairwise correlation fidelity. Those fields are summarized below. DOMIAS, distance-to-closest-record, and alpha-precision were not computed on this panel. Nothing in those fields is a formal privacy guarantee or a HIPAA de-identification claim.

Tree models remain strong tabular predictors, which is why CatBoost is one of the three auditors [35], [36]. TabZilla and TabArena show how brittle a single split can be [37], [38]. The regression lineages come from PMLB [39], [40]. Comparing several methods on the same lineages needs a rank test and a critical difference, or a Bayesian signed-rank alternative [41], [42]. This paper reports the frequentist procedure: a Friedman test and a Nemenyi critical difference on the common set, plus paired Wilcoxon tests. The Bayesian signed-rank procedure is not computed.

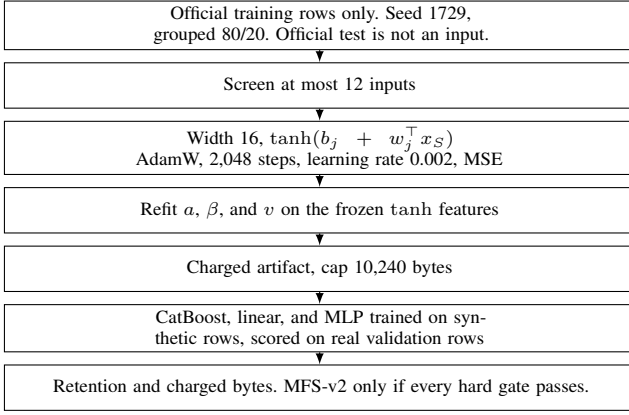


Fig. 1. Displayed pipeline. The official test is not an input.

### III. METHOD

#### A. Displayed generator

The displayed profile is `features12_steps2048`. It is a compact neural residual, not a GAN, a diffusion model, or a transformer. Fig. 1 is the pipeline. Training screens at most 12 inputs. A hidden layer of width 16 maps them through  $\tanh$ . AdamW, learning rate 0.002, runs for 2,048 steps on the regression mean-squared error, with gradient clipping at 5. The hidden weights and biases are then frozen. The intercept, the linear skip, and the hidden readout are refit on those frozen features. Coefficients whose absolute value is below  $10^{-5}$  are dropped from the skip. For a selected input vector  $x_S$ ,

$$h_j = \tanh(b_j + w_j^\top x_S), \quad \hat{y} = a + \beta^\top x_S + v^\top h. \quad (1)$$

Sampling adds the noise model stored in the artifact. There is no embedding table in (1). A separate dataset-action router uses a vector of dimension  $856 + 24 \times (128 + 10) = 4,168$ . That vector does not generate the rows in this paper.

The profile was fixed before the shared retention was inspected. Compile-time search records the proxy  $0.45u + 0.25d + 0.30j$  and then keeps the candidate with the smallest normalized gate shortfall, then the smallest artifact, then the lowest candidate id. Every such compile is marked `production_evidence_unmeasured`. No global DOPE family has been selected. Other shapes exist and are not this result: micro-TVAEs, TinyMAT, joint TVAE, a masked transformer, and a TabSyn candidate. A direct-rank TabDDPM candidate is a research comparator with no claim on the 10,240-byte tier. The scored campaign did not store a per-step loss. A later logged replay did. Section VI plots that replay. The replay files are not substituted for the scored artifacts, and a lower hidden-basis loss at 8,192 steps does not replace the displayed profile. The full width-and-step grid is not measured.

#### B. Retention and MFS-v2

Let  $L_{\text{null}}$ ,  $L_{\text{real}}$ , and  $L_{\text{synthetic}}$  be the auditor's loss on real validation rows when that auditor is trained on a null baseline, on real fit rows, and on synthetic rows. Retention is

$$\frac{L_{\text{null}} - L_{\text{synthetic}}}{L_{\text{null}} - L_{\text{real}}}. \quad (2)$$

TABLE I  
ILLUSTRATION OF (3). NOT A MEASUREMENT.

| Gate map             | Scalar |
|----------------------|--------|
| All six gates passed | 82.40  |
| Byte gate failed     | null   |

A value of 1 means the synthetic-trained auditor matched the real-trained auditor. A value above 1 means it beat that auditor. A negative value means it did worse than the null. A lineage is informative when the real improvement is at least 0.01 times the absolute null loss and is not zero. Otherwise the lineage is omitted. It is not entered as zero. The three auditors are CatBoost [36], ordinary linear regression, and an MLP. The lineage score is the median of sample seeds 101, 211, and 307. The panel score is the median of those lineage scores. Intervals below are percentile lineage-bootstrap intervals, 10,000 resamples of lineages, with the three seeds kept inside the lineage median. They are not per-seed whiskers.

MFS-v2 is a different object. With component  $c_i \in [0, 1]$ , weight  $w_i > 0$ ,  $\epsilon = 10^{-6}$ , and  $W = \sum_i w_i$ ,

$$\text{MFS-v2} = 100 \exp\left(\frac{1}{W} \sum_i w_i \ln(\epsilon + c_i)\right) \quad (3)$$

when the hard-gate map is non-empty, every hard gate passed, and every positive-weight component is finite and inside  $[0, 1]$ . Otherwise the scalar is null. The weights are utility transfer 0.30, driver fidelity 0.20, distribution fidelity 0.20, structure fidelity 0.15, coverage realism 0.10, and compactness 0.05. A privacy failure removes the score through a gate. Utility transfer inside (3) is a clamped one-sided lower bound, not the median of (2). Compactness is  $\text{clamp}(1 - \text{bytes}/10240, 0, 1)$ . Crossing 10,240 bytes fails the byte gate, so the scalar stays null. An artifact over the cap is a failed gate, not a retention of zero.

Table I applies (3) to six invented components (0.90, 0.80, 0.85, 0.70, 0.75, 0.95). The resulting 82.40 is arithmetic, not a measurement. The second row fails the byte gate and the implementation returns null. No artifact in this paper has a measured MFS-v2 value.

### IV. EXPERIMENTAL SETUP

#### A. Partitions and provenance

The panel is the rights-cleared PMLB regression collection [39], [40]. The data lock admits 104 MIT-licensed entries and prepares 100. 4 lineages have no split object because the same source-row digest appears in two partitions (`check_nonoverlap` raises on that overlap). Another 3,755 catalog entries lack a redistribution grant and are not backfilled. The prepared cut is a grouped 80/20 of the official training rows, seed 1729. Worker directories hold `train.csv` and `validation.csv`. They do not hold `test.csv`. Official tests stay sealed, and final evaluation is not authorized. Table II is the lock and the file-name count. Test-file contents were not read.

Fit seed 11 is the population actually measured. Sizes  $n$  and  $4n$  are both stored. The tables below are  $4n$ . The Gaussian

TABLE II  
TEST-SPLIT PROVENANCE. FILE-NAME COUNTS ONLY.

| Item                                               | Record                                                                 |
|----------------------------------------------------|------------------------------------------------------------------------|
| Catalog SHA-256                                    | ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120       |
| Eligible / prepared / overlap exclusions           | 104 / 100 / 4                                                          |
| Unknown-rights catalog entries held out            | 3755                                                                   |
| Split                                              | official_test_grouped_training_80_20, seed 1729                        |
| Overlap lineage ids                                | 590210721973399a, 9fe58a656fdb458a, d06a584f07d0acd5, d0ba629ed07b5d63 |
| Final evaluation authorized                        | false                                                                  |
| S3 worker directories / train.csv / validation.csv | 100 / 100 / 100                                                        |
| test.csv under S3 workers                          | 0                                                                      |
| BeyondArena workers / worker test.csv              | 5 / 0                                                                  |
| Evaluator test.csv files, contents not read        | 5                                                                      |

copula row is Copulas 0.14.1 GaussianMultivariate, selected by held-out mean log density. CTGAN and TVAE are the author implementation, selected by the mean of linear and MLP regression  $R^2$  on validation rows. Chow–Liu uses eight bins and Laplace smoothing 1.0 in study code. Independent marginals are a study reference. The ARF watch log’s last count is 799 closed and 799 ok out of 800 planned, then a sealing record. That log is not a retention panel, and ARF was not sampled on this schedule. Forest-Flow on 6 lineages uses the author regression objective on its frozen grid. Native objectives are not ranked across methods.

### B. Statistics

Comparisons are paired on the lineages where both methods have a complete informative group. The tests are two-sided Wilcoxon signed-rank tests on the nonzero paired differences, and one-sided sign tests for a positive difference, with a matched-pairs rank-biserial correlation. Holm correction is applied inside a family, separately for each test, and is not pooled across families. On CatBoost the Holm sign-test values are  $2.15 \times 10^{-15}$ ,  $6.72 \times 10^{-24}$ , and  $4.95 \times 10^{-27}$ . The density family has nine tests (three comparators times three auditors). The neural family is CTGAN and TVAE times the auditors. The Forest-Flow family is separate and has six lineages. A Friedman test and a Nemenyi critical difference use the four density methods on their common complete-informative set, with rank 1 assigned to the higher retention. The critical-difference constant for four methods at  $\alpha = 0.05$  is 2.569. These are descriptive validation tests. They are not the production paired-superiority rule, which also requires a frozen evaluator and a gate-pass analysis. PTF-v1, MFS-v2, and release-safe L3 stay null.

### C. Compute

Table III is wall time taken from receipts, plus the Forest-Flow artifact byte span. The logged replay is a fresh hidden-basis run of both step budgets. It is not the elapsed time of the original scored campaign, which is not measured on the displayed cells. Forest-Flow fit time is the sum of `fit_seconds` over the 12 hash-checked `fit.json` files. The sample row is the sum of elapsed times in the receipt stored beside each of those files. A shared-evaluator total in the confirmation JSON is not that sum, so it is not reported. The confirmation’s whole-device energy estimate includes idle draw and is not reported as job energy. BeyondArena times are the median `elapsed_seconds` of the 5 prepared receipts.

TABLE III  
SECONDS, EXCEPT THE ARTIFACT ROWS.

| Item                            | Amount        |
|---------------------------------|---------------|
| Replay, 2,048 steps, median     | 4.56          |
| Replay, 8,192 steps, median     | 14.4          |
| Forest-Flow fit core            | 953           |
| Forest-Flow sample and auditors | 8,801         |
| Forest-Flow artifact low        | 115,202.253   |
| Forest-Flow artifact high       | 1,085,155.062 |
| BeyondArena fits, 2,048, median | 5.00          |
| BeyondArena fits, 8,192, median | 14.0          |
| Original displayed campaign     | not measured  |

## V. RESULTS

### A. Density block

Fig. 2 plots retention against  $\log_{10}$  charged bytes for the three auditors. The open markers are each method’s own median and lineage-bootstrap interval on the lineages that method completed. The table intervals are the paired subsets. For CatBoost those two DOPE sets coincide. Paired differences outside  $[-1.5, 1.5]$  are counted in the panel title and omitted from that frame. The retention axis stops at 1.6. The dashed line is the cap. Neural and Forest-Flow artifacts are not on this figure.

Table IV is the paired density block. Under CatBoost, DOPE’s median paired differences are positive against the Gaussian copula (0.224, interval 0.149 to 0.290), the study Chow–Liu tree, and the study independent marginals. The Holm-adjusted  $p$ -values are  $1.35 \times 10^{-13}$ ,  $4.85 \times 10^{-16}$ , and  $1.17 \times 10^{-16}$ . The rank-biserial correlations are 0.890, 0.978, and 0.999. On the 97 lineages common to all four methods, the Friedman  $p$ -value is  $2.71 \times 10^{-44}$  and the critical difference is 0.476. All six CatBoost pairs exceed that difference. Fig. 3 shows the sorted paired differences and that critical-difference scale. The bar is a scale drawn from rank 1. It is not a bracket around one pair.

The linear auditor does not separate DOPE from the Gaussian copula. The average ranks are 1.533 and 1.598, against a critical difference of 0.489. The Holm  $p$ -value is 0.373, the median paired difference is 0.00366 with interval  $-0.00438$  to 0.012, and the rank-biserial correlation is 0.107. The interval contains zero. Chow–Liu and the independent marginals remain separated from DOPE under the linear auditor and under the MLP. The MLP Friedman test separates all six pairs, including the Gaussian copula. A CatBoost separation is not a separation under every auditor.

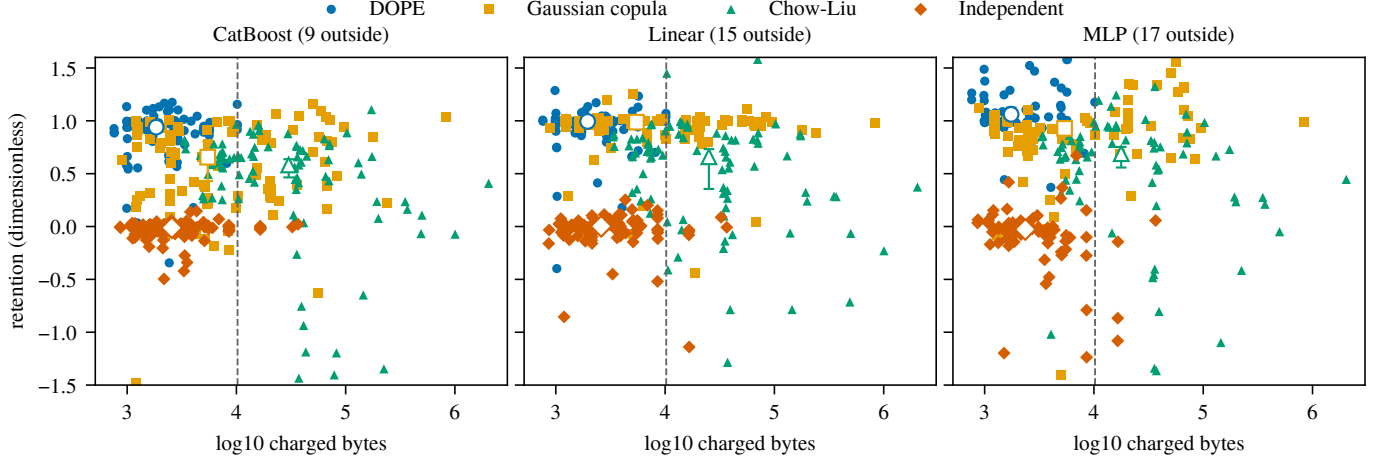


Fig. 2. Density-block retention at  $4n$  against charged bytes. Three auditors. The dashed line is the 10,240-byte cap.

TABLE IV  
DENSITY BLOCK AT  $4n$ . PAIRED MEDIANS AND HOLM  $p$ -VALUES.

| Auditor  | Comparator              | $n$ | DOPE                 | Other                     | Difference                | Holm $p$               |
|----------|-------------------------|-----|----------------------|---------------------------|---------------------------|------------------------|
| CatBoost | Gaussian copula         | 97  | 0.940 [0.914, 0.976] | 0.656 [0.431, 0.803]      | 0.224 [0.149, 0.290]      | $1.35 \times 10^{-13}$ |
| CatBoost | Chow-Liu, study         | 97  | 0.940 [0.914, 0.976] | 0.586 [0.465, 0.637]      | 0.337 [0.282, 0.451]      | $4.85 \times 10^{-16}$ |
| CatBoost | Indep. marginals, study | 97  | 0.940 [0.914, 0.976] | -0.012 [-0.020, -0.00649] | 0.956 [0.918, 0.980]      | $1.17 \times 10^{-16}$ |
| Linear   | Gaussian copula         | 92  | 0.994 [0.981, 0.996] | 0.989 [0.968, 0.994]      | 0.00366 [-0.00438, 0.012] | 0.373                  |
| Linear   | Chow-Liu, study         | 92  | 0.994 [0.981, 0.996] | 0.661 [0.356, 0.738]      | 0.340 [0.244, 0.425]      | $7.12 \times 10^{-15}$ |
| Linear   | Indep. marginals, study | 92  | 0.994 [0.981, 0.996] | 0.00206 [-0.023, 0.00583] | 0.988 [0.975, 1.001]      | $5.70 \times 10^{-16}$ |
| MLP      | Gaussian copula         | 75  | 1.061 [1.024, 1.125] | 0.928 [0.865, 0.953]      | 0.185 [0.128, 0.228]      | $4.96 \times 10^{-11}$ |
| MLP      | Chow-Liu, study         | 75  | 1.061 [1.024, 1.125] | 0.692 [0.582, 0.754]      | 0.530 [0.350, 0.651]      | $2.11 \times 10^{-13}$ |
| MLP      | Indep. marginals, study | 75  | 1.061 [1.024, 1.125] | -0.028 [-0.047, -0.013]   | 1.104 [1.047, 1.179]      | $2.11 \times 10^{-13}$ |

TABLE V  
CHARGED BYTES AT  $4n$ . FOREST-FLOW IS NOT POOLED.

| Method                  | Median bytes  | Within cap |
|-------------------------|---------------|------------|
| DOPE reference          | 1,895.5       | 98/100     |
| Gaussian copula         | 5,464.5       | 62/100     |
| Chow-Liu, study         | 29,865        | 31/100     |
| Indep. marginals, study | 2,549.5       | 94/100     |
| CTGAN                   | 741,834       | 0/22       |
| TVAE                    | 317,141.5     | 0/22       |
| Forest-Flow             | 325,848,913.5 | 0/6        |

Table V counts artifacts at or under the cap. The two displayed DOPE artifacts over the cap are 4544\_Geographical OriginalofMusic at 22,233 bytes and 588\_fri\_c4\_1000\_100 at 19,019 bytes. 678\_visualizing\_environmental has a fit whose CatBoost group is not informative, so the CatBoost median uses 97 lineages while the byte count uses 100. Independent marginals pass the cap more often than the study tree and still sit near retention zero. Smallness alone is not the result. The displayed byte range runs from 758 to 22,233.

On the density cells, the displayed profile has median classifier two-sample AUC 0.517 (interval 0.504 to 0.532,  $n = 98$ ), mean marginal Kolmogorov-Smirnov distance 0.114 (0.103 to 0.135), and pairwise correlation fidelity 0.953 (0.945 to 0.961). A two-sample AUC near one half means a linear-time classifier has trouble telling the synthetic rows from the real validation rows. That is a fidelity description. DOMIAS, distance-to-closest-record, and alpha-precision are

not measured.

### B. Neural block

Table VI is not pooled with Table IV. On the 21 CatBoost lineages, DOPE's median is 0.953 (interval 0.896 to 0.990), CTGAN's is  $-0.081$ , and TVAE's is 0.669. The same direction appears for the linear auditor and, on a smaller common set, for the MLP. Every successful CTGAN and TVAE artifact in this byte count is over the cap (0/22 and 0/22). The native ledger records scheduling cutoffs for the unfinished search. Those cutoffs are unfinished work, not method failures, and they are why this block cannot support a method-wide conclusion.

### C. Forest-Flow

Table VII is the six-lineage confirmation. It is not pooled with the 97-lineage panel or with the neural block. The Forest-Flow CatBoost median is 1.003 (interval 0.926 to 1.026) and the paired DOPE median is 0.940. Charged Forest-Flow artifacts run from 115,202,253 to 1,085,155,062 bytes, with median 325,848,913.5, against a DOPE median of 2,067 bytes on the same six lineages. The utility sits in the same region as DOPE. The byte charge is about five orders of magnitude larger. One row of the table has higher DOPE retention than Forest-Flow. With 6 lineages this block does not support a method-wide ranking. The default and native-selected Forest-Flow charges are the same selection, recorded once. MFS-v2 on this block is null. The campaign is not marked complete.

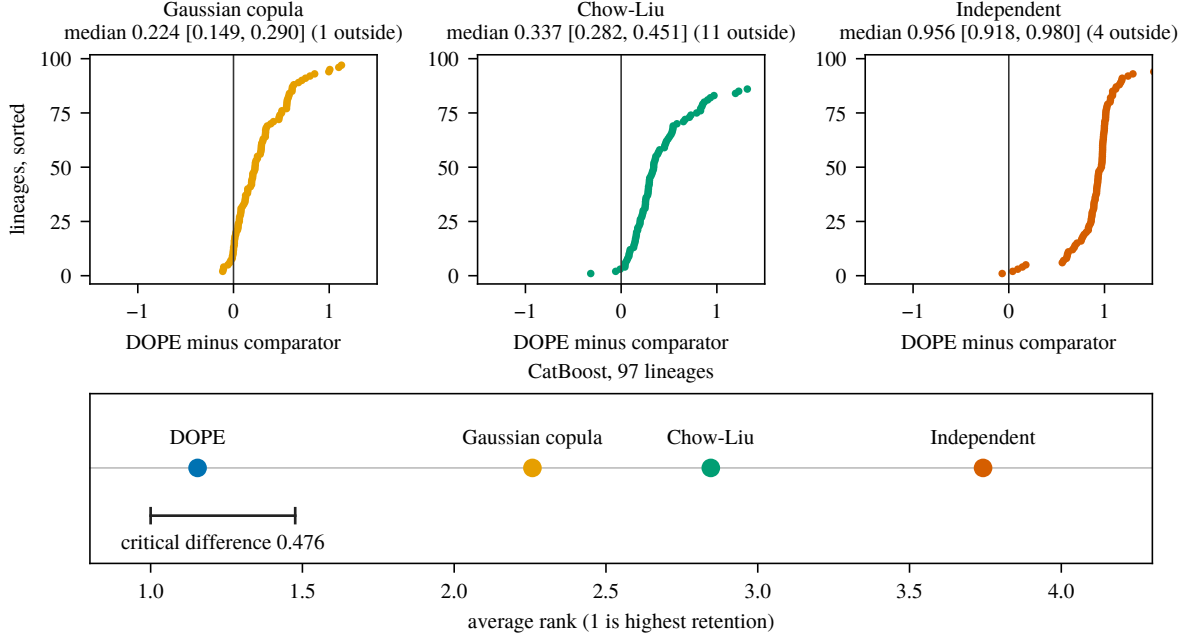


Fig. 3. CatBoost paired differences, sorted, and the Nemenyi scale. Rank 1 is the higher retention. The bar length is the critical difference.

TABLE VI  
NEURAL BLOCK AT  $4n$ . NOT THE DENSITY LINEAGE SET.

| Auditor  | Comparator | $n$ | DOPE                 | Other                  | Difference | Holm $p$              |
|----------|------------|-----|----------------------|------------------------|------------|-----------------------|
| CatBoost | CTGAN      | 21  | 0.953 [0.896, 0.990] | -0.081 [-0.202, 0.103] | 0.881      | $5.72 \times 10^{-6}$ |
| CatBoost | TVAE       | 21  | 0.953 [0.896, 0.990] | 0.669 [0.539, 0.787]   | 0.236      | $1.57 \times 10^{-4}$ |
| Linear   | CTGAN      | 21  | 0.999 [0.996, 1.015] | -0.033 [-0.225, 0.080] | 1.049      | $5.72 \times 10^{-6}$ |
| Linear   | TVAE       | 21  | 0.999 [0.996, 1.015] | 0.796 [0.605, 0.867]   | 0.263      | $1.57 \times 10^{-4}$ |
| MLP      | CTGAN      | 19  | 1.026 [0.998, 1.263] | -0.149 [-0.471, 0.083] | 1.079      | $1.53 \times 10^{-5}$ |
| MLP      | TVAE       | 19  | 1.026 [0.998, 1.263] | 0.753 [0.478, 0.852]   | 0.284      | $1.57 \times 10^{-4}$ |

TABLE VII  
FOREST-FLOW CONFIRMATION AT  $4n$ , CATBOOST. NOT POOLED.

| Lineage           | DOPE bytes | DOPE  | Forest-Flow bytes | Forest-Flow |
|-------------------|------------|-------|-------------------|-------------|
| 1193_BNG_lowbwt   | 2,436      | 0.970 | 416,921,273       | 1.004       |
| 503_wind          | 3,846      | 0.996 | 637,056,689       | 1.016       |
| 582_fri_c1_500_25 | 5,670      | 0.909 | 1,085,155,062     | 0.883       |
| 596_fri_c2_250_5  | 1,698      | 0.807 | 234,776,554       | 0.968       |
| feynman_l12_l     | 1,032      | 0.991 | 123,784,681       | 1.001       |
| strogatz_vdp1     | 984        | 0.581 | 115,202,253       | 1.036       |

#### D. MFS-v2 column

Table VIII is the scalar required by (3). No method has a complete component vector and a fully passed hard-gate map, so every entry is null. Copying a retention number into this column would publish a value the contract refuses to emit. PTF-v1 is null. Release-safe certification is null. The admission locks are not frozen. TabSyn is not measured. TabDDPM on this 100-lineage panel is not measured.

#### E. BeyondArena

BeyondArena is the pinned snapshot

2ecfe882ccfb814fc27c4de10a64ceefd5d7655c.

It is a separate corpus. It is not one of the 100 PMLB lineages, and it is not pooled with them or with the neural

TABLE VIII  
MFS-v2 AFTER NATIVE SELECTION. NULL MEANS NOT EMITTED.

| Method                       | MFS-v2       |
|------------------------------|--------------|
| DOPE reference               | null         |
| Gaussian copula              | null         |
| CTGAN                        | null         |
| TVAE                         | null         |
| Forest-Flow                  | null         |
| ARF                          | null         |
| TabDDPM                      | null         |
| TabSyn                       | not measured |
| synthpop CART                | null         |
| Chow-Liu, study code         | null         |
| Indep. marginals, study code | null         |

block. 5 of 12 locked families were prepared and fit. 7 did

TABLE IX  
BEYONDARENA FAMILIES THAT PREPARED. NO RETENTION AUDITOR.

| Family         | Fit rows | 2,048 B | 8,192 B | Val 2,048 | Val 8,192 |
|----------------|----------|---------|---------|-----------|-----------|
| airfoil        | 802      | 804     | 818     | 0.010     | 0.00448   |
| garments       | 939      | 2,317   | 2,372   | 0.028     | 0.025     |
| musk           | 3,738    | 9,273   | 9,350   | 0.146     | 0.090     |
| telemonitoring | 257      | 1,852   | 1,890   | 0.033     | 0.019     |
| video          | 35,918   | 5,054   | 5,071   | 0.00247   | 0.00195   |

TABLE X  
CATBOOST MEDIAN (LINEAGE COUNT) BY INFORMATIVENESS THRESHOLD.

| Threshold | DOPE       | Gaussian   | Chow-Liu   | Indep.      |
|-----------|------------|------------|------------|-------------|
| 0         | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.001     | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.01      | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.05      | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.1       | 0.942 (96) | 0.668 (98) | 0.584 (98) | -0.012 (98) |

not prepare: 4 were outside the projected feature range, and 3 hit an overlapping source row. The range was not raised, and overlapping rows were not dropped to force a fit. The prepared artifacts are inside the cap. Table IX records charged bytes and the final hidden-basis validation loss, before the readout refit. No retention auditor was run, so the block has no retention column. The supplement names all 142 families.

## VI. ABLATIONS

`features12_steps8192` is the same residual with four times as many AdamW steps. It is a budget check, not the displayed model. On the lineages where both budgets have a complete three-seed group, the CatBoost medians are 0.926 at 2,048 steps and 0.970 at 8,192 steps (68 lineages; median paired difference 0.034). The linear medians are 0.990 and 0.987 (64 lineages; difference  $-0.000283$ ). The MLP medians are 1.065 and 1.057 (52 lineages; difference 0.00451). The supplement lists all 100 lineages at both budgets. A blank cell is an unavailable fit or an auditor that is not informative. It is not a zero.

The scored fits did not keep a per-step series. Fig. 4 is the logged replay of both budgets, written to a new directory and not substituted for the scored artifacts. The curves are the hidden-basis objective before the closed-form readout refit. All 100 lineages finished both budgets. The median training loss starts at 0.321 and ends at 0.00274 after 2,048 steps and at 0.000763 after 8,192 steps. The median validation loss starts at 0.271 and ends at 0.00443 and 0.00181. The band is a 95% lineage-bootstrap interval evaluated every eight steps and filled linearly between those steps. The lower hidden-basis loss at 8,192 steps does not replace the displayed profile.

The primary informativeness threshold is 0.01 times the absolute null loss. On the displayed CatBoost cells that threshold matches the lineage record. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Table X repeats the check for the other density methods. Fit-seed variance across seeds other than 11 is not measured.

## VII. DISCUSSION

The measurements support a limited statement. On this validation panel, the displayed residual, charged at a median of 1,895.5 bytes, recovers a CatBoost median retention of 0.940 and is separated from the three density comparators under that auditor and under the MLP. It is not separated from the Gaussian copula under the linear auditor. Forest-Flow, on 6 confirmation lineages, reaches similar CatBoost retention at a byte charge five orders of magnitude larger. CTGAN and TVAE, on their own 21-lineage block, do not match that retention and do not fit under the cap in the successful size-4 artifacts. None of those statements is an MFS-v2 score, a PTF-v1 result, or a release-safe certification. The production scalar stays null until every hard gate passes and every component is present. A small Wilcoxon  $p$ -value on retention does not flip that null.

The 8,192-step budget moves the CatBoost median by 0.034 on the paired subset and does not become the displayed model. BeyondArena shows that 5 further families fit inside the cap. It does not show retention. TabSyn, a 100-lineage TabDDPM panel, an equal documented tuning budget, and a privacy or detection attack are not measured. The supplement carries the lineage rows that used to sit in the main article.

## VIII. THREATS TO VALIDITY

*a) Internal.:* The fits use coresets, one fit seed, and a training-derived validation cut of the official training rows. The official test was not an input to training, selection, or the numbers in this paper. The informativeness rule drops lineages whose real-versus-null gap is negligible. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. A different threshold can change the median’s denominator. Incomplete groups are omitted rather than scored as zero, which raises the median relative to a policy that would impute failures as zero. Chow-Liu and independent marginals are study implementations, not the authors’ packages. CTGAN and TVAE are native-selected on the lineages that finished. Unfinished scheduling cutoffs are not evidence that those methods fail.

*b) External.:* These fits do not establish behavior on the full tables used by the diffusion papers, on a five-seed campaign, or on the official test. Generalization past seed 1729 is not measured. Forest-Flow’s byte comparison uses 6 lineages and the hash-checked fit files. BeyondArena is a separate fit check. It is not a retention panel, and it is not pooled with the density block.

*c) Construct.:* Retention is the reported utility. It is not the MFS-v2 scalar, which stays null, so this paper does not rank methods by that scalar. The linear auditor does not separate DOPE from the Gaussian copula. A CatBoost separation is not a separation under every auditor. Fidelity fields already on the density cells are not a membership-inference evaluation. Empirical resemblance is not differential privacy and is not HIPAA de-identification.

*d) Statistical.:* The Friedman and Wilcoxon results are descriptive. They are not the pre-registered production superiority test, because that test also requires gate-pass outcomes and a frozen evaluator. The four blocks answer different questions.

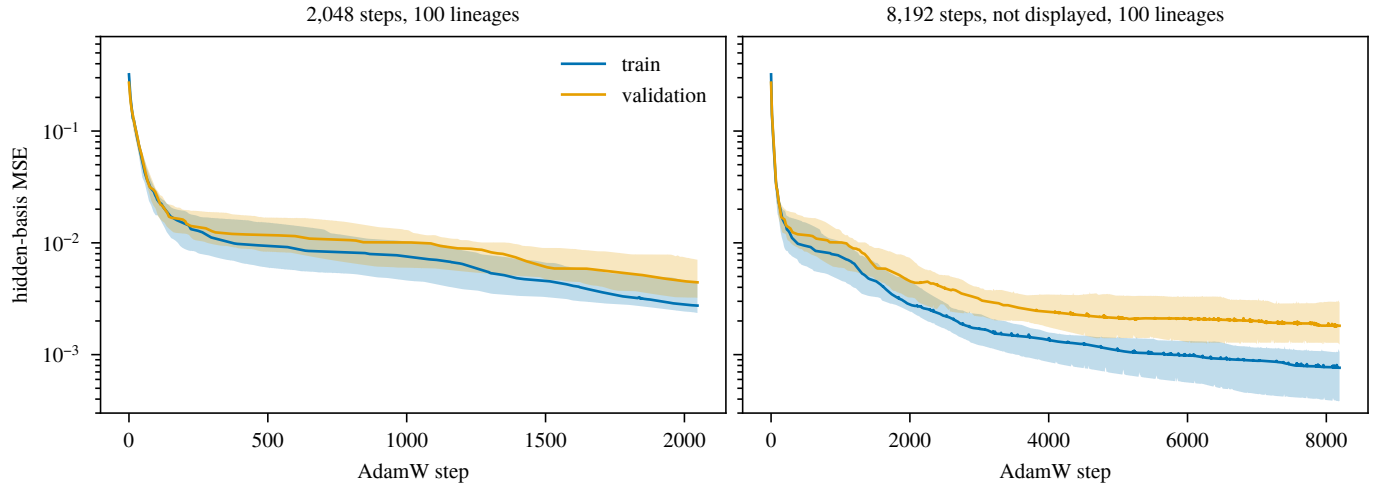


Fig. 4. Hidden-basis MSE on a log axis, before the readout refit. The 8,192-step panel is not the displayed model. The band is a lineage-bootstrap interval.

Pooling the 97-lineage density medians with the 21-lineage neural medians, the 6-lineage Forest-Flow medians, or the 5 BeyondArena losses would invent a sample size the ledgers do not have. A small  $p$ -value on retention does not flip the null certification scalar.

## IX. CONCLUSION

`features12_steps2048` is a 10,240-byte residual. On 97 validation lineages its CatBoost retention is 0.940 (interval 0.914 to 0.976), and 98/100 artifacts fall inside the cap. Paired tests separate it from the Gaussian copula, a study Chow–Liu tree, and study independent marginals under CatBoost, and they do not separate it from the Gaussian copula under the linear auditor. Forest-Flow reaches similar CatBoost retention only at a far larger byte charge, on 6 lineages. The common certification scalar is null, the official tests are sealed, and the generators that were not run are not measured.

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## APPENDIX A REPRODUCIBILITY

Regenerate the tables, the three figure PDFs, and this PDF with just paper from the commit that contains this file. The generator reads the committed validation ledgers, the hash-checked Forest-Flow `fit.json` files, the logged-replay receipts, and the ARF watch log. It does not open an official test file. The catalog hash is `ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120`. The grouped training split uses seed 1729. The measured fit seed is 11, and the sample seeds are 101, 211, and 307. `python3 research/benchmark/verify_paper_numbers.py` exits nonzero when a displayed macro disagrees with those files. Hardware for the Forest-Flow confirmation fits was the GPU host named in the fit files. Replay and BeyondArena wall times are the receipt field `elapsed_seconds`. Placeholder cells for TabSyn, a full TabDDPM panel, privacy attacks, fit-seed variance, and the full ablation grid stay the words “not measured” until those runs exist.